

NBA Projections - Lever Rundown

Plain-English summary of every architectural lever tried (2026-04-29 / 04-30 sessions)

v4-lite scaffold

TQ + gravity teammate effects

How: Added teammate-quality + gravity covariates to the per-game NB2 model.
Result: [WIN] Shipped to Wonka. Captures pace + role context cleanly.

young-on-tank + alpha_promotion

How: Per-position promotion-rate covariates to capture role expansion on bad teams.
Result: [LOSS] Removed in cross-season ablation - no consistent signal, added noise.

Avenue 4 - per-position phi

Per-position NB2 overdispersion

How: Replaced scalar phi with a per-position vector so guards/forwards/centers each get their own variance calibration.
Result: [WIN] 11-stat overnight sweep clean, coverage_80 in [0.77, 0.86] - tight to nominal.

Shooting-percentage attempts

Joint LKJ 3PT/FT correlation

How: Cholesky-factored covariance on player random effects to share information across stats.
Result: [LOSS] Hierarchical-pool collapse - all elite shooters projected near league mean.

Independent 3PT% Stan

How: Same hierarchical scaffold without the LKJ joint structure, looser priors.
Result: [LOSS] Same collapse - confirmed cause was not the LKJ piece.

EB Beta-Binomial 3PT% with FT% transfer

How: Closed-form conjugate update: each player's prior 3PT% set from their FT% via a transfer function, then updated by their attempts.
Result: [WIN] Spread fully recovered: Curry projected 0.405 vs Stan's collapsed 0.36.

EB-rate * v4-lite FG3A volume

How: Multiplied the clean EB rate by v4-lite's projected per-36 FG3A.
Result: [LOSS] v4-lite FG3A is bad enough that combined output was 17% worse than modeling makes directly.

EB-rate * oracle FG3A (sanity)

How: Multiplied EB rate by ACTUAL test-season FG3A to isolate rate quality from volume quality.
Result: [WIN] 46% improvement - confirms rate generalizes; bottleneck is volume side.

FG3A volume covariates

team_3PA_rate as single covariate

How: Added prior-season team three-point attempt rate as a covariate to v4-lite FG3A.
Result: [LOSS] Only 3.7% variance explained among stayers, ~0% among movers - too weak to overcome noise.

Minutes (mpg) projection

Hierarchical Gaussian on log(mpg)

How: Stan model with age curve, team_changed flag, prior-mpg regression, rookie indicator.
Result: [LOSS] Collapsed catastrophically - all 641 players projected within 1.5 mpg, KD = 17 mpg vs reality 37.

Mode-count diagnostic on mpg/48

How: KDE peak detection on the empirical distribution before choosing model form.
Result: [WIN] Found 3 modes (bench/rotation/star) - confirms unimodal won't work, mixture model needed.

Turnovers

Own-usage z-score covariate

How: Added beta_own_usage to v3 Stan to capture within-player usage shifts driving turnovers.
Result: [LOSS] Identifiable with mu_player intercept (R-hat 1.044 on the lift), MAE got worse.

Opponent strength (Lever 2)

Season-level opp_strength_z

How: Minutes-weighted opponent defensive rating as a per-row covariate.
Result: [LOSS] Signal too small at season-level granularity (averages out).

Ship raw def_rating to Wonka

How: Exported team_def_rating_history.parquet for week-level use downstream.
Result: [WIN] Wonka uses it for week-level matchup adjustments where the signal lives.

Pre-fit safety

Empirical-vs-prior calibration ratio

How: Diagnostic that compares empirical player-level spread to the prior's expected spread before any Stan fit.
Result: [WIN] Caught what would've been the 4th over-shrinkage before any compute. Now standard practice.

Shot-zone v5 (Stage 1)

Per-zone NB2 volume

How: Decomposed FGA into 6 zones (RA / paint / mid / lc3 / rc3 / ab3); hierarchical pooling per zone.
Result: [WIN] All zones passed pre-fit calibration, fit landed in 16 min, per-zone phi excellent (RA=17, Paint=10).

Shot-zone v5 (Stage 2)

Per-zone binomial-logit efficiency

How: Player-level FG% by zone with hierarchical pooling, composes with Stage 1 volume.
Result: [WIN] FG-only MAE 0.98 PTS/36 vs 2.38 with league-average FG% - 59% reduction. Stage 1+2 reconstruction in healthy band.

Shot-zone v5 (Stage 3-4)

Role-slot projection blending (fixed alpha heuristic)

How: When a player moves teams, blend their intrinsic zone allocation with the role's expected per-36 zone volumes (departing-player avg for trades, own prior per-36 for stayers, team avg for FAs). $\alpha=1$ stayer / 0.5 mover / 0.3 FA.
Result: [WIN] Modest 2.8% MAE improvement on movers+FAs, but big wins on trade-to-smaller-role: Jrue Holiday MIL->BOS error reduced by 3.34 FGA, Bradley Beal WAS->PHX by 3.59. Stayer-breakouts still missed (Brunson).

Shot-zone v5 (Stage 5)

Hierarchical alpha learned from history

How: Per-player alpha estimated inside Stan with partial pooling toward position then league mean. Goal: replace the fixed-alpha heuristic with data-driven estimates.
Result: [LOSS] alpha collapsed to ~0.98 for everyone ($\kappa_{\alpha}=138$, very tight pool). Identifiability gap predicted in design notes: 46% stayer rows give alpha no traction, only 21% trade rows can't override the position pool. Stage 3-4 v1 fixed-alpha heuristic wins.

v6 drive-volume / FTA (v1, multiplicative)

Multiplicative compose drives + postups

How: Projected per-36 FTA = $\text{drives_per_36} * \text{fta_per_drive_eb} + \text{postup_per_36} * \text{fta_per_postup_eb} * 1.10$ residual.
Result: [LOSS] Drives + post-ups cover only 33% of total FTA; residual factor varies hugely by player ($q_{25}=2.18$, $q_{75}=4.37$). Empirical median residual is 2.95, not 1.10. Even with calibrated factor still loses to v4-lite.

v6 v2 (direct NB2 with covariates)

Direct NB2 with drives + paint + postup as per-36 covariates

How: Reframed as regression: model FTA directly with drives, paint_touches, post_touches as centered per-36 covariates. Lets Stan learn the linear combination.
Result: [WIN] MAE 0.71 beats both naive (0.79, +9.9%) and v4-lite (0.73, +1.9%). Per-position phi 16-22 vs v4-lite's 2.4 - 5-10x precision improvement. The architectural fix the queue review predicted.

Mpg projection redo

3-component Gaussian mixture on log(mpg)

How: Bench/rotation/star components with covariate-dependent mixing weights. Built after mode diagnostic detected 3 modes.
Result: [LOSS] MAE 5.37 vs naive 3.68 (-46%). Weighted-average projection compresses extremes (KD: 37 actual, 23 projected). Mixture obscures year-over-year autocorrelation that naive captures cleanly.

Simple Gaussian regression on log(mpg)

How: OLS regression with $\log_{\text{prior_mpg}} + \text{age} + \text{team_changed} + \text{position dummies}$, no per-player intercept, no mixture.
Result: [LOSS] MAE 3.82 vs naive 3.68 (-3.8%). Coefficients sane ($\beta_{\text{prior}}=0.79$ mean reversion, $\beta_{\text{team_changed}}=-0.06$) but breakouts (Cam Thomas 22->31, Coby White 24->36) and role-loss cases (Bullock 30->9) unpredictable from public data. Mpg projection is genuinely hard at this granularity; naive is the ship target.

Defensive STL/BLK diagnostic

Tracking covariates (deflections, rim_dfg_pct) for STL/BLK

How: Tested whether prior-season tracking metrics explain v4-lite STL/BLK residuals on team-switchers and role-shifters.
Result: [MIXED] BLK: prior_blk alone has $r=0.88$ with future BLK, fully absorbs the signal. Tracking adds nothing (residual $r < 0.10$). STL: deflections has marginal signal (partial $r=0.21$ controlling for prior_stl). BLK stays on v4-lite scaffold.

STL with deflections covariate (build attempt)

How: Built season-level NB2 on STL with deflections_per36 as centered covariate, mirroring v6 v2 architecture.
Result: [LOSS] MAE 0.221 - beat naive +9% but lost to v4-lite -16.6% (0.189). Partial $r=0.21$ wasn't enough to overcome the architecture penalty of season-level vs v4-lite's game-level NB2. v4-lite STL stays as is.

Shot-zone v5 stayer-breakout absorption

Teammate-departure absorption for stayers

How: When a stayer's team had significant departures ($\geq 10\%$ of retained minutes), blend their intrinsic with departing players' avg per-36 zone profile, scaled by stayer's minutes share. Cap at 30% volume increase.
Result: [PARTIAL WIN] Brunson NYK->NYK projection improved by +2.90 FGA (15.3->18.2 vs actual 21.7). Stable stayers (KAT) stay neutral. Toronto cohort (Siakam, Anunoby) over-absorbed slightly because not all retained players absorb proportionally to minutes.

AST + tracking covariates (passing endpoint)

Direct NB2 with passes_made, potential_ast, secondary_ast, drive_ast

How: Same architecture as v6 v2 FTA but for AST. Tracking covariates have strong effects ($\text{potential_ast} +0.107$, $\text{secondary_ast} +0.111$).
Result: [LOSS] MAE 0.744 vs v4-lite 0.666 (-11.7%). Same architecture penalty as STL deflections - season-level NB2 with covariates loses to v4-lite's game-level scaffold when v4-lite is already well-specified.

TOV + tracking covariates (drive_tov)

Direct NB2 with drives, drive_tov, post_touch_tov

How: Same architecture as v6 v2 FTA. Tracking shows huge effects (beta_drive_tov +0.62, beta_post_touch_tov +0.84).

Result: [LOSS] MAE 0.363 vs v4-lite 0.314 (-15.6%). Confirms architectural rule: covariate-augmented season models only beat v4-lite where v4-lite is structurally underspecified (FTA had phi=2.4, AST/TOV/STL/REB do not).

Lever 2 (FT% Bayesian shrinkage diagnostic)

Decompose FTM error into FTA-side vs FT%-side

How: Substitute actual FTA $\times$ proj FT% and proj FTA $\times$ actual FT% to isolate which side carries FTM error.

Result: [DISCONFIRMED] FTA error 4x larger than FT% error (0.547 vs 0.129). EB FT% modeling would deliver ~0 lift. Saves ~1h of build time.

FTM compose ship

FTM = v6 v2 FTA $\times$ Laplace-smoothed train FT%

How: Multiply v6 v2 FTA projection by career-train FT% with Laplace prior of 50 attempts at league mean (0.78).

Result: [WIN] MAE 0.583 - beats v4-lite by 3.7% and naive by 10.0%. Has +0.157 bias from FTA over-projection on low-volume players (hierarchical pooling regression-toward-mean). Ship as FTM override.

Lever 3 (game-script intentional foul)

Team close-game share $\times$ low-FT% interaction as FTA covariate

How: Built team-level close-game frequency from PBP (margin ≤ 5 in last 3 min of 4th Q). Added intentional_foul_proxy = max(0, 0.78 - career_ft) $\times$ close_game_share as 4th covariate to v6 v2 FTA model.

Result: [LOSS] MAE 0.713 vs v6 v2 0.712 (-0.1%). Coefficient came back NEGATIVE (-0.92) - model learned 'low-FT% bigs benched in close games' not 'hacked.' Cheap team-level proxy can't substitute for lineup-level minutes-by-margin engine. Real Hack-a-Shaq mechanism requires player-presence-during-close-late which PBPStats provides; raw nba_api would be 2-3 weeks engineering for ~1.4% lift on 34-player cohort.

Architectural lessons (the wins from the losses)

1. Hierarchical pooling collapses when player obs are sparse and recency-weighted - use empirical-Bayes conjugate updates for rate stats, not Stan hierarchical pooling.
2. Aggregate covariate fixes (team_3PA_rate, own_usage) don't beat structural factorization. When a stat is downstream of a more fundamental quantity, model the upstream quantity.
3. Multiplicative composition needs ALL event sources or it systematically undershoots. Direct NB2 regression with all sources as covariates beats compose-then-multiply (v6 v2 vs v6 v1).
4. Pre-fit calibration ratio (empirical SD vs prior expected SD, target [0.5, 2.0]) is now standard practice. It caught what would've been the 4th hierarchical-pool collapse before any compute.
5. When fixing one layer surfaces error in an adjacent layer, that's correct progress. Each iteration approaches correct factorization one layer at a time.
6. Mixture models on heavily autocorrelated stats (mpg) can lose to naive 'predict next = prior'. The mixture's weighted-average projection compresses extremes while naive preserves them. Sometimes the simpler baseline is just genuinely hard to beat.
7. Hierarchical alpha (per-player learned blending parameter) is unidentifiable when most training data is from stable role contexts. Fixed-alpha heuristics can outperform learned alpha when role-shift training rows are sparse. Validates the design-time skepticism about identifiability.
8. Fast cheap diagnostics beat fast cheap fits. The defensive STL/BLK diagnostic showed prior-stat history fully absorbs tracking signal for BLK, marginally for STL - saved a 3-5h fit cycle that would have shipped a model with no real lift.